

Aayush Jaiswal

aaajais@iu.edu | linkedin.com/in/jaiswal-aayush | github.com/aayush9400 | [Google Scholar](https://scholar.google.com/citations?user=aaajais) | medium.com/@jaayush12

SUMMARY

Applied Scientist with 3+ years of expertise in **machine learning research** and **model training optimization**, specializing in efficient model architectures and scalable AI systems. Proven track record in developing **language models**, generative models, and production ML systems with focus on **model compression**, fine-tuning, and **AI safety**. Passionate about building interpretable, high-performance AI solutions that balance efficiency with real-world impact.

SKILLS SUMMARY

Proficient in: Data Engineering, Machine Learning, Deep Learning, Generative Models (TensorFlow, PyTorch)

Cloud Platforms: AWS, GCP, with MLOps practices

Data Pipelines: Spark, DBT, Airflow

Research: Generative models, diffusion, VAEs

Advanced ML Techniques: Production use-cases, high performance, reliability

EXPERIENCE

AI Solutions Engineer

Feb 2026 – Present

CoreWeave

Post-Sales

- Enabled enterprise customers to deploy efficient models by integrating leading MLOps solutions.

AI Support Engineer

Oct 2024 – Feb 2026

Weights & Biases

San Francisco, CA

- Delivered scalable training pipelines using PyTorch, TensorFlow, Docker, Kubernetes across AWS/GCP, cutting workflow overhead by 20%
- Optimized distributed training for cross-functional teams, resolving model bottlenecks and boosting GPU utilization by 30%
- Evaluated emerging frameworks to accelerate AI workflows and enhance model interpretability

AI Researcher

Aug 2023 – Oct 2024

Indiana University

Bloomington, IN

- Achieved SOTA 3D generative models via compression and distributed training on HPC clusters
- Integrated AI safety protocols and evaluation frameworks, improving model reliability and interpretability
- Mentored 150+ students on scalable ML pipelines and cloud deployment best practices

Machine Learning Engineer

Jan 2021 – Mar 2022

Trell (Sequoia-backed)

Remote

- Drove 25% retention and 45% watch-time gains through transfer learning and fine-tuned personalization models
- Scaled data pipelines with Spark/DBT for millions of daily records, cutting training time by 30%
- Achieved 99.9% uptime via real-time monitoring, reducing debugging time by 30%
- Built MIS dashboard integrating ERP & CRM data to power data-informed product decisions

Data Science Intern

Jun 2019 – Jul 2019

Airtel

Delhi, India

- Segmented 10M+ users via clustering, boosting marketing ROI by 10%
- Orchestrated data pipelines processing 10M+ daily records with 99% quality assurance

TECHNICAL SKILLS

ML Frameworks & Tools: PyTorch, TensorFlow, JAX, Scikit-Learn, HuggingFace Transformers, LangChain, MLflow, W&B

Language Models & NLP: Fine-tuning, Prompt Engineering, Transfer Learning, Model Compression, Domain Adaptation, RAG Pipelines

ML Techniques: Generative Models (VAE, GAN, Diffusion), Computer Vision, Recommendation Systems, Reinforcement Learning

Training & Infrastructure: Distributed Training, Model Optimization, Parameter Tuning, Scalable AI Pipelines, Data Pipeline Design

Cloud & MLOps: AWS, GCP, Azure, Docker, Kubernetes, Spark, dbt, Airflow, Datadog, Prometheus

Programming: Python (Advanced), SQL (Advanced), Java (Intermediate), C/C++ (Intermediate)

AI Safety & Ethics: Model Interpretability, Fairness in AI, Alignment Principles, Robustness Testing

KEY PROJECTS

Computer Vision Model | *YOLOv8, OpenCV, Python, AWS*

- Developed real-time object detection reducing manual review by 85%
- Processed 50K+ images/month with 98% detection accuracy

RAG Chatbot with LangChain | *LangChain, Pinecone, OpenAI, FastAPI*

- Built production RAG system serving 10K+ queries/day with 95% accuracy
- Reduced response latency by 60% through vector optimization and caching

Meal Planner AI Agent | *Flutter, FastAPI, Firebase, AWS, OpenAI, LangChain, MongoDB, Docker, Fine-tuning, RAG*

- Developed a smart meal planning assistant using **fine-tuned language models** and **prompt engineering techniques** to generate personalized meal recommendations aligned with user preferences and dietary constraints.
- Implemented **RAG (Retrieval-Augmented Generation) pipelines** with semantic search and knowledge graphs to enhance LLM response accuracy and relevance based on calendar schedules and meal timing patterns.
- Optimized **model inference efficiency** by **30%** through prompt optimization and caching strategies, reducing API costs while maintaining response quality and user engagement.
- Deployed full-stack production system integrating LLM capabilities with user-centered design principles, ensuring **model safety** and appropriate content generation.

3D Image Conditional Diffusion Model | *Python, PyTorch, TensorFlow, HPC, Distributed Training, W&B, Model Compression*

- Designed a novel 2-channel VQ-VAE **model compression architecture**, achieving **state-of-the-art** SSIM and PSNR metrics while reducing model size by **60%** for efficient deployment.
- Developed and evaluated **training methodologies** for generative diffusion models on 3D medical imaging data, implementing **distributed training** strategies on HPC infrastructure to scale model training.
- Optimized **model training pipelines** through hyperparameter tuning and architectural improvements, reducing training time by **40%** while improving model robustness and generalization.
- Applied rigorous evaluation frameworks to ensure model safety, reliability, and clinical applicability of synthetic medical image generation.

Multi-Lingual Short Video Recommendation System | *TensorFlow, AWS, GCP, SQL, Docker, MLFlow, CI/CD, Transfer Learning*

- Architected a custom recommendation system using **transfer learning** and **domain adaptation techniques**, increasing retrieval speed by **50%** and improving content relevance across multiple languages.
- Implemented automated **model retraining pipeline** using AWS Lambda and MLFlow, effectively managing data drift and ensuring model performance consistency over time.
- Designed **scalable training infrastructure** supporting continuous model updates and A/B testing frameworks to evaluate model improvements in production environments.

Monitoring for Cloud-Native Environments | *AWS, Prometheus, Grafana, Kubernetes, PromQL, S3*

- Implemented scalable monitoring pipeline on AWS, enhancing system reliability, uptime, and performance insights by over 30%.

Real-Time Data Pipeline Using Spark and DBT | *Python, Spark, Airflow, GCP*

- Improved data analytics efficiency by 15% through real-time data pipelines using Spark and orchestrated workflows with Airflow.
- Ensured data reliability and consistency across distributed cloud systems.

PUBLICATIONS

Classification of the COVID-19 Infected Patients Using DenseNet201 Based Deep Transfer Learning | *Taylor & Francis*

- Achieved 96.25% accuracy and AUC of 0.97 in classifying COVID-19 from chest CT scans using DenseNet201 and deep transfer learning techniques.
- Significantly contributed to rapid diagnosis techniques, demonstrating the real-world impact of AI in healthcare.

Rapid COVID-19 Diagnosis Using Ensemble Deep Transfer Learning Models from Chest Radiographic Images | *Taylor & Francis*

- Designed ensemble models achieving 98.25% accuracy, enhancing the effectiveness of large-scale COVID-19 screening programs.
- Demonstrated proficiency in developing advanced model ensembles with significant societal impact.

EDUCATION

Indiana University Bloomington

Master of Science in Data Science

Manipal University Jaipur

Bachelor of Technology in Computer Science

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